Independent $O(\alpha_s)$ Calculation for the B_c Axial Current
Lecture Notes, Validation Tests, and Current Audit Status

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Abstract

These notes describe the momentum-space calculation of the perturbative $O(\alpha_s)$ correction to the axial-current correlator used in the B_c mixing analysis. The purpose is twofold: to give a derivation that a graduate student can reproduce and to expose every convention needed for an independent theoretical check. The leading-order, soft, ultraviolet-pole, infrared-pole, equal-mass form-factor, scale-cancellation, and small-mass checks pass. Two normalization errors in an earlier provisional result are identified explicitly: the Package-X epsilon-bar symbol was inverted, and the $O(\epsilon)$ difference between the D-dimensional and four-dimensional projected Born traces was omitted. After correcting both, the independent axial coefficient reproduces the known equal-mass result and has a smooth massless limit. Numerical results are quoted separately in an on-shell diagnostic convention and in a common-scale $\overline{\text{MS}}$ convention.

Contents

1 Physics goal and conventions

The axial current is

$$J_\mu^A(x) = \bar{b}(x)\gamma_\mu\gamma_5 c(x), \tag{1}$$

and the two-point function is

$$\Pi_{\mu\nu}^{AA}(p) = i\int d^4x\, e^{ip\cdot x}\langle 0|T\{J_\mu^A(x)J_\nu^{A\dagger}(0)\}|0\rangle. \tag{2}$$

We decompose

$$\Pi_{\mu\nu}^{AA}(p) = \left(g_{\mu\nu} - \frac{p_\mu p_\nu}{p^2}\right)\Pi_1^{AA}(p^2) + \frac{p_\mu p_\nu}{p^2}\Pi_0^{AA}(p^2) \tag{3}$$

and project with

$$P_1^{\mu\nu} = \frac{1}{3}\left(g^{\mu\nu} - \frac{p^\mu p^\nu}{p^2}\right) \tag{4}$$

in four dimensions. In the virtual loop, the projector must instead be

$$P_{1,D}^{\mu\nu} = \frac{1}{D-1}\left(g_D^{\mu\nu} - \frac{p^\mu p^\nu}{p^2}\right), \qquad D = 4-2\epsilon. \tag{5}$$

The perturbative spectral density is defined by

$$\rho_{\text{pert}}^{AA}(s) = \rho_0^{AA}(s) + \frac{\alpha_s}{\pi}\rho_1^{AA}(s) + O(\alpha_s^2). \tag{6}$$

Thus ρ_1 never contains the explicit factor α_s/π.

2 Leading-order cut

For the cut state $c(p_c)\bar{b}(p_b)$, $p = p_c + p_b$, the spin sum is

$$T_{\mu\nu}^{(0)} = \mathrm{Tr}\left[(\not{p}_c + m_c)\gamma_\mu\gamma_5(\not{p}_b - m_b)\gamma_\nu\gamma_5\right]. \tag{7}$$

The on-shell relations are

$$p_c^2 = m_c^2, \qquad p_b^2 = m_b^2, \qquad 2p_c\cdot p_b = s - m_c^2 - m_b^2. \tag{8}$$

The two-body phase-space factor is proportional to

$$\frac{\sqrt{\lambda(s,m_b^2,m_c^2)}}{s}, \qquad \lambda(s,m_b^2,m_c^2) = [s-(m_b+m_c)^2][s-(m_b-m_c)^2]. \tag{9}$$

In the normalization of the momentum-space mixing code,

$$\rho_0^{AA}(s) = \frac{N_c}{16\pi^2}\frac{\sqrt{\lambda}}{s}\mathcal{N}_{AA}(s). \tag{10}$$

The independently evaluated trace agrees algebraically with `PerturbativeSpectralDensity["AA",s]`:

```
IndependentAALODerivationCheck[s]["Difference"]
(* 0 *)
```

3 NLO decomposition

The inclusive correction is divided into

$$\rho_1^{AA} = \rho_{1,V}^{AA} + \rho_{1,R-D}^{AA} + \rho_{1,I}^{AA}, \tag{11}$$

where V is the renormalized virtual correction, $R-D$ is the exact three-body real contribution minus local dipoles, and I is the integrated dipole insertion. The individual terms depend on subtraction conventions; only their sum is physical.

4 Virtual correction

4.1 Loop amplitude

With loop momentum ℓ, the vertex denominators are represented by

$$\frac{1}{\ell^2\left[(\ell+p_c)^2-m_c^2\right]\left[(\ell-p_b)^2-m_b^2\right]}. \tag{12}$$

The numerator is traced in D dimensions. In FeynCalc this requires `GAD`, `GSD`, `MTD`, and `FVD`; performing a four-dimensional trace and applying `ChangeDimension` afterward is not equivalent because $O(\epsilon)$ numerator terms can multiply $1/\epsilon$ poles.

There are two nonsinglet γ_5 matrices. They are anticommuted together before the trace is continued to D dimensions. Thus no four-dimensional γ_5 object is mixed with a D-dimensional gamma chain. The corresponding four-dimensional identity is checked by

```
IndependentAALOGamma5EliminationCheck[]
(* 0 *)
```

The tensor integrals are reduced with

$$\texttt{TID} \longrightarrow \texttt{ToPaVe} \longrightarrow \texttt{PaXEvaluateUVIRSplit}. \tag{13}$$

The physical loop measure $d^D\ell/(2\pi)^D$ is supplied through `PaXImplicitPrefactor`.

4.2 Epsilon-bar convention and projected Born normalization

FeynHelpers defines

$$\frac{1}{\texttt{PaXEpsilonBar}} = \frac{1}{\epsilon} - \gamma_E + \ln(4\pi). \tag{14}$$

Therefore the Laurent-series substitution is `PaXEpsilonBar` $\to z$, not $1/z$. Reversing this substitution was the dominant error in the superseded +3% result.

The second necessary correction follows from the projected Born trace. The ratio between the D-dimensional gamma-five-free trace and the four-dimensional trace is

$$\frac{B_D}{B_4} = 1 + \epsilon\,\frac{2[(m_b-m_c)^2-s]}{3(m_b-m_c)^2+6s} + O(\epsilon^2). \tag{15}$$

For equal masses this becomes $1-\epsilon/3$. Since the virtual graph contains $1/\epsilon$ poles, this apparently evanescent factor changes the finite result. The loop is consequently multiplied by B_4/B_D before the finite part is extracted.

4.3 On-shell field counterterm

For the coefficient of $\alpha_s/(4\pi)$,

$$\delta Z_{2,Q}^{\mathrm{OS}} = -C_F\left[\frac{1}{\epsilon_{\mathrm{UV}}} + \frac{2}{\epsilon_{\mathrm{IR}}} + 4 + 3\ln\frac{\mu^2}{m_Q^2}\right]. \tag{16}$$

The interference of the two external-field factors with the Born amplitude gives

$$\rho_{1,\mathrm{field}}^{AA} = \frac{1}{4}\left(\delta Z_{2,b}^{\mathrm{OS}} + \delta Z_{2,c}^{\mathrm{OS}}\right)\rho_0^{AA}. \tag{17}$$

The factor $1/4$ converts the $\alpha_s/(4\pi)$ convention into the α_s/π convention.

5 Real-gluon emission

For

$$J_A \longrightarrow c(p_c) + \bar{b}(p_b) + g(k), \tag{18}$$

define

$$u = 2p_c\cdot k, \qquad v = 2p_b\cdot k, \qquad t = (p_c+p_b)^2 = s-u-v. \tag{19}$$

The two emission chains are

$$\mathcal{M}_c^{\alpha} = \gamma^{\alpha}\frac{\not{p}_c + \not{k} + m_c}{u}\gamma_\mu\gamma_5, \tag{20}$$

$$\mathcal{M}_{\bar{b}}^{\alpha} = \gamma_\mu\gamma_5\frac{-\not{p}_b - \not{k} + m_b}{v}\gamma^{\alpha}. \tag{21}$$

Their sum obeys the soft theorem. The three-body measure after integrating the overall orientation is

$$d\Phi_3 = \frac{dt\,du}{128\pi^3 s}. \tag{22}$$

The spectral density includes the optical-theorem factor $1/(2\pi)$.

At fixed t,

$$u_\pm(t) = \frac{s-t}{2t}\left[t + m_c^2 - m_b^2 \pm \sqrt{\lambda(t, m_b^2, m_c^2)}\right]. \tag{23}$$

6 Local massive dipoles

For a massive emitter Q, emitted gluon g, and massive spectator k, the Catani–Dittmaier–Seymour–Trocsanyi dipole has the form

$$D_{gQ,k} = \frac{1}{2p_g\cdot p_Q}\langle V_{gQ,k}\rangle|\mathcal{M}_0(\widetilde{p}_Q, \widetilde{p}_k)|^2. \tag{24}$$

The local subtraction was checked analytically:

```
IndependentAADipoleSoftCancellationCheck[s,yAA]["CancelsQ"]
(* True *)
```

This is necessary but not sufficient: the quasi-collinear and integrated finite limits must also be checked.

7 Complete integrated insertion

A key lesson from the audit is that the individual integrated dipole $I_{gQ,k}$ is not by itself the complete finite insertion. For final-state partons, the insertion operator is

$$\mathbf{I}_m(\epsilon) = -\frac{\alpha_s}{2\pi}\frac{(4\pi)^\epsilon}{\Gamma(1-\epsilon)}\sum_j \frac{1}{\mathbf{T}_j^2}\sum_{k\neq j}\mathbf{T}_j\cdot\mathbf{T}_k\left[\mathbf{T}_j^2\left(\frac{\mu^2}{s_{jk}}\right)^\epsilon\left(\mathcal{V}_j-\frac{\pi^2}{3}\right)\right.$$
$$\left. +\Gamma_j+\gamma_j\ln\frac{\mu^2}{s_{jk}}+\gamma_j+K_j\right]. \tag{25}$$

For a quark,

$$\gamma_q = \frac{3}{2}C_F, \qquad K_q = \left(\frac{7}{2}-\frac{\pi^2}{6}\right)C_F, \tag{26}$$

and for a massive quark,

$$\Gamma_q(\mu, m_q; \epsilon) = C_F\left[\frac{1}{\epsilon}+\frac{1}{2}\ln\frac{m_q^2}{\mu^2}-2\right]. \tag{27}$$

The functions $\mathcal{V}^{(S)}$ and $\mathcal{V}_q^{(NS)}$ are implemented from Eqs. (6.20) and (6.21) of Catani et al., Nucl. Phys. B627 (2002) 189.

8 Pole and scale checks

The following tests currently pass:

1. The LO cut reproduces the stored spectral density.
2. The exact real matrix element reproduces the universal massive soft eikonal factor.
3. The virtual UV pole cancels the on-shell field counterterm.
4. The real and virtual IR-pole coefficients cancel.
5. The derivative of the finite virtual term with respect to $\ln\mu^2$ equals its IR-pole coefficient.

The scale-slope test is particularly useful because it checks the dimensional prefactor without depending on a published full spectral density.

9 How to validate a large correction

A large correction is not disproved by its size alone. Near a heavy-heavy threshold, gluon exchange is enhanced by the small relative velocity. Nevertheless, a large result requires more tests than a small one. It is useful to organize them in three levels.

9.1 Level I: algebraic and numerical identities

These tests use no external spectral-density formula:

1. exchange the two masses, $m_b \leftrightarrow m_c$;
2. rescale every dimensional variable;
3. change the numerical quadrature order;

4. vary the subtraction cutoff or integration variables;
5. verify cancellation of the renormalization scale in the complete on-shell coefficient.

The AA density has mass dimension two. Therefore

$$\rho_1(\kappa^2 s, \kappa m_b, \kappa m_c, \kappa\mu) = \kappa^2 \rho_1(s, m_b, m_c, \mu). \tag{28}$$

For $\kappa = 2$, the numerical ratio between the left-hand side and the right-hand side is 0.9999988. Exchanging m_b and m_c changes $\rho_1(s = 40\,\mathrm{GeV}^2)$ by only 1.8×10^{-14}.

9.2 Level II: limits derived directly from QCD

Threshold or NRQCD limit. For two heavy quarks the appropriate effective description near threshold is NRQCD rather than ordinary heavy-light HQET. Define

$$v_{\mathrm{rel}} = \frac{\sqrt{\lambda(s, m_b^2, m_c^2)}}{s - m_b^2 - m_c^2}. \tag{29}$$

Single Coulomb-gluon exchange implies

$$\frac{\rho_1}{\rho_0} = \frac{C_F \pi^2}{v_{\mathrm{rel}}} + O(\ln v_{\mathrm{rel}}, 1). \tag{30}$$

This is the first term in the expansion of the Sommerfeld/Coulomb factor. For the physical mass inputs the numerical check is

$s - s_{\mathrm{th}}$ [GeV^2]	v_{rel}	ρ_1/ρ_0	$v_{\mathrm{rel}}\rho_1/\rho_0$
0.10	0.13629	95.1736	12.9711
0.25	0.21326	60.1795	12.8340
0.50	0.29653	42.6488	12.6465
1.00	0.40589	30.3493	12.3186

The limiting expectation is $C_F\pi^2 = 13.1595$. The approach toward this number is a strong explanation for the large Borel correction: the Borel weight emphasizes the near-threshold region. It also warns that fixed-order perturbation theory may eventually require Coulomb resummation.

High-energy limit. For $s \gg m_b^2, m_c^2$, both masses become negligible and the nonsinglet axial result approaches the massless value

$$\frac{\rho_1}{\rho_0} \longrightarrow 1. \tag{31}$$

For the physical masses, the calculated ratios are

$$\begin{array}{c|ccc} s\,[\mathrm{GeV}^2] & 100 & 400 & 1600 \\ \hline \rho_1/\rho_0 & 4.0352 & 1.9458 & 1.3035 \end{array}. \tag{32}$$

The convergence is slow because the bottom mass is not negligible at the first two points, but the sequence approaches one.

Heavy-light limit. Sending one mass to zero while keeping the other fixed provides a different check. Ordinary HQET becomes relevant near the heavy-light threshold. At $s = 40\,\mathrm{GeV}^2$, $m_b = 4.18\,\mathrm{GeV}$, and $m_c = 1.0, 0.5, 0.2, 0.1, 0.05\,\mathrm{GeV}$, the ratios ρ_1/ρ_0 are

$$8.4611, \quad 6.4876, \quad 5.6258, \quad 5.3519, \quad 5.2036. \tag{33}$$

The sequence is finite and smooth. A stronger check is to match the exact normalization to the known heavy-light correlator of Chetyrkin and Steinhauser [4]. Because their published basis is not identical to the present projected axial convention, that comparison must be performed at the level of a carefully matched spectral density rather than by comparing one number.

9.3 Level III: independent published calculations

The following comparisons probe different parts of the result:

1. Catani et al. [1]: massive subtraction and the exact equal-mass virtual form factors.
2. Chetyrkin and Steinhauser [4]: the heavy-light, one-massless-quark limit and its HQET threshold expansion.
3. Lee, Sang and Kim [5]: unequal-mass $b\bar{c}$ current matching in NRQCD, including the one-loop and relativistic expansion. Their principal application is to S-wave B_c currents, whereas the transverse axial channel studied here has a P-wave threshold structure. The comparison is therefore limited to carefully matched Lorentz components and short-distance conventions; it is not a direct spectral-density check.
4. Maier and Marquard [6]: high-energy expansions of non-diagonal vector/scalar correlators and diagonal axial correlators. These results test the expected expansion structure, but they are not a published unequal-mass axial formula for the present current.
5. Wang [2]: a direct unequal-mass NLO spectral-density comparison, treated only as an external check because the local transcription still requires an independent convention audit.

9.4 What the old K-factor model checks

The earlier diagnostic replaced

$$\Pi^{ij}_{\mathrm{pert}} \to \left(1 + \frac{\alpha_s}{\pi} K_{ij}\right) \Pi^{ij}_{\mathrm{pert}}. \tag{34}$$

It remains useful for propagating hypothetical channel-dependent corrections through

$$\tan(2\theta) = \frac{-2\Pi^{AB}}{\Pi^{AA} - \Pi^{BB}}. \tag{35}$$

If all K_{ij} are equal, a common correction largely cancels in the angle. If they differ, the angle can move by several degrees. The scan therefore measures the *sensitivity of the angle*; it cannot validate $\rho_1^{AA}(s)$, reproduce threshold logarithms, or assign a statistical NLO uncertainty.

10 Mandatory limiting tests

10.1 Massless limit

For equal masses followed by $m^2/s \to 0$, the transverse nonsinglet axial current becomes the massless vector/axial current. Therefore

$$\frac{\rho_1^{AA}(s)}{\rho_0^{AA}(s)} \longrightarrow 1. \tag{36}$$

At $s = 40\,\mathrm{GeV}^2$, the corrected numerical sequence is

$m_b = m_c$ [GeV]	ρ_1^{AA}/ρ_0^{AA}
0.20	1.0831974
0.10	1.0249053
0.05	1.0072604

which approaches one as required.

10.2 Equal-mass form factor

The virtual contribution is checked against the equal-mass form factors in Appendix D of Catani et al. Both f_1 and the finite-mass pure-axial f_2 term are included. At $s = 40\,\mathrm{GeV}^2$ and $\mu = \sqrt{s}$, the comparison is

m [GeV]	FeynCalc/Package-X	Catani axial	difference
0.20	42.54933479185	42.54933479167	1.75×10^{-10}
0.50	26.78426097716	26.78426097716	-4.71×10^{-12}
1.00	18.02429851421	18.02429851421	2.49×10^{-14}

This validates the finite virtual normalization independently of the unequal-mass literature.

10.3 Unequal-mass external comparison

The expressions of Z.-G. Wang, arXiv:1303.4146, may be used only after:

1. reproducing their LO normalization;
2. checking their equal-mass limit analytically rather than substituting directly into formulas containing $(m_1 - m_2)^{-1}$;
3. checking the massless limit;
4. comparing several values of s, not one integrated number.

The current local transcription of that formula does not pass all these requirements and is not used to construct the result.

11 Borel transformation

Once a validated ρ_1^{AA} is available, its continuum-subtracted Borel moment is

$$\Pi_{1,\mathrm{NLO}}^{AA}(M^2, s_0) = \frac{\alpha_s(\mu)}{\pi} \int_{(m_b+m_c)^2}^{s_0} ds\, e^{-s/M^2} \rho_1^{AA}(s,\mu). \tag{37}$$

For the fixed numerical inputs $m_b = 4.18\,\mathrm{GeV}$, $m_c = 1.27\,\mathrm{GeV}$, $\alpha_s = 0.26$, $M^2 = 10\,\mathrm{GeV}^2$, and $s_0 = 55\,\mathrm{GeV}^2$, interpreted only as an on-shell-scheme diagnostic, the pointwise decomposition at $s = 40\,\mathrm{GeV}^2$ is

$$\rho_{1,V} = 5.6979732, \qquad \rho_{1,I} = -2.2122115, \qquad \rho_{1,R-D} = 0.4477501, \tag{38}$$

so that $\rho_1^{AA} = 3.9335117$. The LO density is $\rho_0^{AA} = 0.3898262$, and the local relative correction $(\alpha_s/\pi)\rho_1/\rho_0$ is 0.8351.

The continuum-subtracted 20-node Borel result is

$$\Pi_{\mathrm{LO}}^{AA} = 0.1441716826, \tag{39}$$

$$\Pi_1^{AA} = 1.3770512496, \tag{40}$$

$$\frac{\alpha_s}{\pi}\Pi_1^{AA} = 0.1139655469, \tag{41}$$

$$\Pi_{\mathrm{LO+NLO}}^{AA} = 0.2581372295. \tag{42}$$

Thus the relative correction is 0.790485, or 79.05%. The 12-node and 20-node bare coefficients are 1.3770552219 and 1.3770512496, respectively. This agreement tests the outer Borel quadrature.

This large correction is not combined with the paper's $\overline{\mathrm{MS}}$ mass inputs, because 4.18 and 1.27 GeV are normally interpreted as $m_b(m_b)$ and $m_c(m_c)$, not pole masses.

12 $\overline{\rm MS}$ conversion

Conversion to running masses uses

$$M_Q = \overline{m}_Q(\mu)\left[1+\frac{\alpha_s(\mu)}{\pi}\left(\frac{4}{3}+\ln\frac{\mu^2}{\overline{m}_Q^2(\mu)}\right)\right], \tag{43}$$

which implies

$$\rho^{AA}_{1,\overline{\rm MS}} = \rho^{AA}_{1,\rm OS} + \sum_{Q=b,c}\overline{m}_Q(\mu)\left(\frac{4}{3}+\ln\frac{\mu^2}{\overline{m}_Q^2(\mu)}\right)\frac{\partial\rho_0^{AA}}{\partial m_Q}. \tag{44}$$

Using $m_b(m_b) = 4.18\,\text{GeV}$, $m_c(m_c) = 1.27\,\text{GeV}$, $\alpha_s(M_Z) = 0.1180$, one-loop running, and the common scale $\mu = 2.0\,\text{GeV}$, the parameters are

$$\overline{m}_b(\mu) = 4.6683009\,\text{GeV}, \qquad \overline{m}_c(\mu) = 1.1659209\,\text{GeV}, \qquad \alpha_s(\mu) = 0.2622100. \tag{45}$$

The corrected 20-node result is

$$\Pi^{AA}_{\rm LO,\overline{MS}} = 0.0723406411, \tag{46}$$

$$\Pi^{AA}_{1,\rm OS}\big|_{\overline{m}(\mu)} = 0.7641393212, \tag{47}$$

$$\Pi^{AA}_{1,\rm conversion} = -0.1431034913, \tag{48}$$

$$\Pi^{AA}_{1,\overline{\rm MS}} = 0.6210358299, \tag{49}$$

$$\frac{\alpha_s(\mu)}{\pi}\Pi^{AA}_{1,\overline{\rm MS}} = 0.0518341572, \tag{50}$$

$$\Pi^{AA}_{\rm LO+NLO,\overline{MS}} = 0.1241747983. \tag{51}$$

The relative NLO correction is 71.65%. Varying the common scale gives

μ [GeV]	$\Pi^{AA}_{\rm LO}$	$(\alpha_s/\pi)\Pi^{AA}_1$	$\Pi^{AA}_{\rm LO+NLO}$
1.8	0.05866679	0.05981541	0.11848220
2.0	0.07234064	0.05183386	0.12417450
2.2	0.08587655	0.04205294	0.12792949

The central value should therefore be quoted with a renormalization-scale band. The fixed-input on-shell and common-scale $\overline{\rm MS}$ numbers answer different scheme questions and must not be added to each other.

13 Reproducible Mathematica audit

```
Get["BcMixingAlphaS.wl"];

IndependentAALODerivationCheck[s]["Difference"]
IndependentAARealEmissionSoftTheoremCheck[s,yAA]["Difference"]
IndependentAAUVPoleCancellationCheck[
  s,l,FeynCalc`PaXC0Expand->True
]["CancelsQ"]
IndependentAAIRPoleCancellationCheck[40.0][
  "CancelsNumericallyQ"
]

IndependentAAEqualMassVirtualBenchmarkReport[
  40,1/5,Sqrt[40]
```

```
]["PassesFiniteBenchmarkQ"]

IndependentAAThresholdCoulombCheck[
  {0.1,0.25,0.5,1.0},
  $BcMixingDefaultParameters,
  AccuracyGoal->5,PrecisionGoal->5,MaxRecursion->10
]

IndependentAAHighEnergyLimitCheck[
  {100.,400.,1600.},
  $BcMixingDefaultParameters,
  AccuracyGoal->5,PrecisionGoal->5,MaxRecursion->10
]

IndependentAAMassExchangeSymmetryCheck[
  40.,$BcMixingDefaultParameters,
  AccuracyGoal->5,PrecisionGoal->5,MaxRecursion->10
]

IndependentAADimensionalScalingCheck[
  40.,2.,$BcMixingDefaultParameters,
  AccuracyGoal->5,PrecisionGoal->5,MaxRecursion->10
]

aaOS = IndependentAAFinalNLOBorelSummary[
  10.0,55.0,$BcMixingDefaultParameters,
  "NPoints"->20,"Progress"->False,
  AccuracyGoal->5,PrecisionGoal->5,MaxRecursion->10
];

aaMSbar = IndependentAAFinalNLOBorelMSbarSummary[
  10.0,55.0,2.0,
  $BcAlphaSMSbarDefaults,$BcMixingDefaultParameters,
  "NPoints"->20,"Progress"->False,
  AccuracyGoal->5,PrecisionGoal->5,MaxRecursion->10
];
```

14 Current conclusion

- The original LO plus G^2 and G^3 mixing analysis is not invalidated by this audit.
- The provisional +3% AA correction is withdrawn. It resulted from reversing the epsilon-bar substitution and omitting the B_4/B_D normalization.
- The corrected independent AA coefficient passes the equal-mass virtual benchmark, ultraviolet and infrared cancellation, scale cancellation, quadrature convergence, and the massless inclusive limit.
- At the test point, the correction is large: approximately 79% in the fixed-input on-shell diagnostic and 72% in the common-scale $\overline{\mathrm{MS}}$ calculation at $\mu = 2$ GeV.
- This is the correction to the AA moment, not yet to the mixing angle. A consistent NLO mixing angle requires independent AB and BB coefficients, where cancellations in the angle formula may reduce or enhance the final shift.

References

[1] S. Catani, S. Dittmaier, M. H. Seymour and Z. Trocsanyi, Nucl. Phys. B **627** (2002) 189, arXiv:hep-ph/0201036.

[2] Z.-G. Wang, *Next-to-leading order perturbative contributions in the QCD sum rules for mesonic two-point correlation functions with unequal quark masses*, arXiv:1303.4146.

[3] K. G. Chetyrkin, J. H. Kühn and M. Steinhauser, Comput. Phys. Commun. **133** (2000) 43, arXiv:hep-ph/0004189.

[4] K. G. Chetyrkin and M. Steinhauser, *Heavy-light current correlators at order α_s^2 in QCD and HQET*, arXiv:hep-ph/0108017.

[5] J. Lee, W. Sang and S. Kim, *Relativistic corrections to the axial vector and vector currents in the $\bar{b}c$ meson system at order α_s*, arXiv:1011.2274.

[6] A. Maier and P. Marquard, *Low- and high-energy expansion of heavy-quark correlators at next-to-next-to-leading order*, arXiv:1110.5581.